

ENTERPRISE APPLICATION SPECIFICATION

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Product Catalog, Technical Specifications, Pricing & Sales Discovery System

Client: SVG Electric & Control Products (India)

DOCUMENT CONTROL INFORMATION

Document Reference:	BRD-SVG-ELE-2026-V1
System Name:	SVG Electric Estimation & Product Management System
Target Audience:	Executive Management, Sales Engineers, Software Engineering Team
Scope Revision:	Version 1.0 (Streamlined Admin Product Entry & Sales Price Finder)
Document Owner:	Principal Solutions Architect & Engineering Lead
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1. EXECUTIVE SUMMARY & PROJECT BACKGROUND

SVG Electric & Control Products (India) is an established industrial manufacturer specializing in Motor Control Centers (MCC), Power Control Centers (PCC), Variable Frequency Drive (VFD) panels, Automation/PLC cabinets, APFC power factor correction capacitor switchboards, and distribution systems.

To accelerate sales cycle velocity and eliminate pricing bottlenecks, SVG Electric commissioned this internal web application. While earlier iterations featured heavy multistage estimation approval workflows, customer CRM databases, and rule matrices, client field operations identified that sales engineers primarily require an instantaneous, high-precision Product Search & Price Discovery Engine. Concurrently, administrative staff require an agile Product Catalog Entry Console where technical parameters, direct final prices, and engineering drawings/PDF datasheets can be entered in minutes.

This BRD establishes the definitive operational, functional, and technical blueprint for the deployed version 1.0 system.

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TABLE OF CONTENTS

SECTION NAME	PAGE
1. Executive Summary & Project Background	Page 1
2. Business Problem & Transformed Scope Analysis	Page 2
3. User Personas & Role-Based Access Control (RBAC)	Page 2
4. Functional Requirements Specification (FR-1 through FR-7)	Page 3
5. Technical Architecture & Technology Stack	Page 4
6. Data Dictionary (Core Product Model)	Page 4
7. API Architecture & Endpoint Contracts	Page 5
8. Non-Functional Requirements (Performance, Security, Reliability)	Page 5
9. Verification & Approval Sign-Off Matrix	Page 5

2. BUSINESS PROBLEM & SCOPE ANALYSIS

Historical electrical estimation workflows involved manual lookups across static spreadsheets, delays in waiting for admin approval, and inability to quickly answer customer technical inquiries regarding specific ratings (e.g. "What is our price for an 800A IP54 MCC panel?").

CAPABILITY AREA	PHASE 1 LEGACY MODEL	PHASE 2 STREAMLINED MODEL (APPROVED)
Pricing Methodology	Complex formula combination matrix	Direct Admin Final Price per product entry
Technical Specs	Static dropdown attributes	Dynamic Key-Value Spec Builder + Quick suggestions
Technical Attachments	None	Direct Image & PDF datasheet upload (<25MB)
Sales Workflow	Draft estimation -> Submit -> Admin Approval	Instant keyword & parameter-based price discovery
Client Submittals	Generic multi-page estimation quote	Branded Technical Spec Sheet with optional price toggle

3. USER PERSONAS & ROLE-BASED ACCESS CONTROL (RBAC)

PERSONA	PRIMARY OBJECTIVES & CAPABILITIES	ASSIGNED ROUTE ACCESS
ADMINISTRATOR (Engineering & Operations Head)	- Create, modify, and delete product panel entries. - Configure dynamic technical specifications (Current, Voltage, IP, Form, Busbar). - Set final commercial prices in Indian Rupees. - Upload panel photographs and PDF engineering drawings. - Manage user logins and audit logs.	/dashboard (Overview) /admin/products /admin/users /admin/settings
SALES ENGINEER (Field Technical Sales)	- Real-time keyword search across panel models and specifications. - Filter panels by category and electrical parameters (e.g. 800A, 415V). - Retrieve exact final prices immediately during client calls. - Preview/download technical PDF datasheets and GA drawings. - Generate printable technical submittals (with or without price).	/dashboard (Product Search & Price Finder)

4. FUNCTIONAL REQUIREMENTS SPECIFICATION

The following functional requirements govern the complete operation of the Version 1.0 system across authentication, catalog management, pricing, files

REQ ID & FEATURE	PRIORITY	DETAILED FUNCTIONAL REQUIREMENT
FR-1: Authentication & RBAC	High	The system shall authenticate users via email and bcrypt-hashed password, issuing secure HTTP-only JWT cookies. Middleware guards must intercept and enforce role boundaries (e.g. restricting /admin routes to ADMIN role).
FR-2: Product Master Entry & Specs Builder	High	Admin shall create products with Product Code (SKU), Name, Category, and Description. An interactive Specifications Builder must allow arbitrary Key-Value technical parameters with one-click electrical suggestions (Rated Current, Operating Voltage, Ingress Protection, Busbar Material, Form of Separation, Incomer Switchgear).
FR-3: Direct Final Commercial Pricing	High	Admin shall directly specify the final ex-works unit price in INR for each product entry. The system must format this into Indian numbering standard (e.g., Rs. 3,85,000) and store it in the database.
FR-4: Multi-Format Attachment Pipeline	High	The system shall provide an asynchronous upload API (/api/upload) supporting images (.jpg, .jpeg, .png, .webp) and PDF documents (.pdf) up to 25MB. Files must be sanitized, assigned timestamped unique filenames, and stored in public/uploads/ with full preview and download capabilities.
FR-5: Sales Search & Filter Engine	High	Sales engineers shall search products in real-time. The query parser must search across product codes, product names, categories, descriptions, and dynamic JSON specification values. Parameter-specific dropdowns must allow filtering by exact ratings.
FR-6: Dual-Mode Catalog Presentation	Medium	The UI shall support instantaneous switching between Visual Grid Cards (featuring visual thumbnails, specs badges, and prominent price tags) and Dense Technical Tables (optimized for rapid multi-item comparisons).
FR-7: Printable Technical Spec Sheet	High	The system must render a publication-grade technical datasheet modal. A prominent "Show Price on Print" toggle must allow users to include or omit commercial pricing. When printing via window.print(), background navigation must hide, leaving a clean A4 technical submittal.

5. TECHNICAL SYSTEM ARCHITECTURE & TECHNOLOGY STACK

LAYER / COMPONENT	SELECTED TECHNOLOGY	ARCHITECTURAL RATIONALE
Frontend Framework	Next.js 14 (App Router) + React 18	Server-Side Rendering, fast client hydration, robust route nesting
Styling & Design System	Tailwind CSS + Lucide Icons	Industrial design system, consistent brand palette, print stylesheets
Backend API Layer	Next.js Route Handlers (Node.js)	Secure RESTful endpoints, streaming multipart file uploads
ORM & Database Layer	Prisma ORM + PostgreSQL	Type-safe schema, relational integrity, native JSON column support
Authentication	JWT (jose) + Bcrypt.js	Stateless session tokens, HTTP-only cookie security, salted hashing

6. DATABASE SCHEMA & DATA DICTIONARY

FIELD NAME	TYPE	CONSTRAINTS	BUSINESS PURPOSE
id	String	PK, UUID	Unique immutable product identifier
productCode	String	Unique, Indexed	Commercial model SKU (e.g. MCC-IND-01)
name	String	Required	Full descriptive product title
price	Float	Default 0.0	Final ex-works commercial unit price in INR
categoryId	String	FK -> Category	Relational link to panel category
specifications	Json	Nullable, Array	Structured list of [{ name: string, value: string }]
fileUrl	String	Nullable	Path to uploaded drawing/datasheet in /uploads/
fileName	String	Nullable	Original user file name with extension
fileType	String	IMAGE PDF	MIME classification for appropriate UI rendering
active	Boolean	Default true	Controls sales catalog search visibility

7. API ARCHITECTURE & ENDPOINT CONTRACTS

ENDPOINT	METHOD	AUTH ROLE	PAYLOAD & RESPONSE SUMMARY
/api/auth/login	POST	Public	Validates credentials; sets HTTP-only auth_session cookie
/api/auth/logout	POST	All	Invalidates session cookie; redirects to /login
/api/upload	POST	Authenticated	Multipart form data (file); returns {fileUrl, fileName, fileType, fileSize}
/api/admin/products	GET, POST, PUT, DELETE	ADMIN	Full CRUD on product entries with specs, price, and attachments
/api/products/search	GET	SALES / ADMIN	Searches by keyword q, categoryId, specName, specValue; returns matching products

8. NON-FUNCTIONAL REQUIREMENTS (NFR)

DIMENSION	REQUIREMENT & ARCHITECTURE GUARANTEE
Performance	Search queries execute in < 80ms across the full panel catalog. Assets load with cache-control headers.
Security	HTTP-only secure JWT cookies prevent XSS. Password hashes use 10-round bcrypt salt. Sanitized file uploads.
Reliability	PostgreSQL transactional ACID guarantees. Zero data loss on concurrent price updates.
Usability	Industrial dark sidebar with ElectCare logo. Responsive layout across desktop workstations and mobile tablets.

9. VERIFICATION & APPROVAL SIGN-OFF MATRIX

ROLE / STAKEHOLDER	SIGN-OFF NAME	VERIFICATION STATUS	DATE
Client Business Sponsor	SVG Management	[APPROVED] / [NEED REVISION]	
Lead Systems Architect	Principal Solution Architect	[APPROVED] / [NEED REVISION]	